

SCHOOL STUDENT PORTAL SYSTEM

Group 3

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1. INTRODUCTION (STORY-BASED OPENING)

In many schools today, students struggle with accessing their academic results on time. Lecturers also face challenges when managing large numbers of student grades manually, and administrators often find it difficult to track tuition fees and ensure that only eligible students receive academic privileges.

This project was born from that real-life challenge.

Imagine a student named "Alex," who has completed all his examinations and is waiting eagerly to see his grades. At the same time, the lecturer has already prepared the results, but the administrator must first confirm whether Alex has completed his fee payments before the grades are released. This simple but powerful idea inspired the creation of the School Student Portal System — a digital platform that connects students, lecturers, and administrators in one controlled and intelligent system. It is not just a project. It is a solution designed to bring fairness, order, and structure into academic management.

2. BACKGROUND OF THE PROJECT

The system was developed using a structured programming approach through VS Code. It represents a web-based student portal designed to manage academic and financial activities in a school environment.

The system integrates three main users:

Administrator, Lecturer, and Student.

Each user has a unique role and cannot access another user's functions without permission.

3. PROBLEM STATEMENT

Before this system, schools faced several challenges such as:

Manual handling of student records

Difficulty in tracking tuition fee payments

Delay in releasing student grades

Risk of unauthorized access to academic results

Lack of coordination between lecturers and administration

Repetition or duplication of student registration

These issues often caused confusion, delays, and unfair access to academic information.

4. AIM AND OBJECTIVES

Aim: To design and develop a secure, structured, and efficient student portal system that manages academic records and fee control using role-based access.

Objectives: To create a digital student registration system

To allow lecturers to input and manage student grades

To ensure administrators control fee verification and approval

To restrict grade access based on payment status

To prevent duplicate student registrations

To improve communication between students, lecturers, and administration

5. SYSTEM OVERVIEW

The School Student Portal System is a role-based web application that manages academic and financial data.

It operates through three dashboards:

Admin Dashboard

Lecturer Dashboard

Student Dashboard

Each dashboard is protected by login authentication and role permissions.

6. USER ROLES AND RESPONSIBILITIES

ADMINISTRATOR MODULE

The administrator has full control over financial and approval processes

Responsibilities:

Manage student accounts

Verify and track tuition fee payments

Approve or block access to grades

Confirm student eligibility before results are released

Maintain financial records

Key Role:

The administrator acts as a gatekeeper between academic results and student access.

LECTURER MODULE

The lecturer is responsible for academic assessment.

Responsibilities:

Access lecturer dashboard

Search students using unique student ID or Moodle code

Enter and submit grades

Send grade data for administrative approval

Key Role:

Lecturers prepare and submit results but cannot directly release them.

STUDENT MODULE

The student is the end user of the system.

Responsibilities:

Register and log in

Access personal dashboard

Request academic results and View fee status

Access grades when approved

Key Role:

Students only gain access to results after full system approval.

7. SYSTEM WORKFLOW (HOW IT WORKS STEP-BY-STEP)

Step 1: Registration

Students enter: Name, Student ID, Course and Faculty

The system ensures each ID is unique and prevents duplication.

Step 2: Login System

Users log in based on roles:

Admin, Lecturer and Student

Each role opens a different dashboard.

Step 3: Grade Entry by Lecturer

Lecturer searches student using ID or Moodle code

Lecturer enters grades

Grades are forwarded to the admin system

Step 4: Administrative Verification

Admin checks fee payment status

If fees are incomplete → grades are blocked

If fees are complete → approval is granted

Step 5: Grade Release

Once approved: Student can view grades

Dashboard displays:

Name, ID, Course and Grades

Step 6: Fee Payment System

Students pay fees through the system

Receipt is generated, and Payment status is updated automatically

8. KEY SYSTEM FEATURES

- Role-Based Access Control (RBAC)
- Secure login authentication
- Fee verification system
- Grade approval workflow
- Unique student ID validation
- Dashboard-based interface
- Notification system
- Data protection and integrity
- Prevention of duplicate registration

9. SYSTEM ARCHITECTURE CONCEPT

- The system follows a layered structure:
- Frontend (User Interface)
- Backend (Logic Processing)
- Database (Data Storage)
- Each layer works together to ensure smooth operation of the portal.

10. IMPORTANCE OF THE SYSTEM

- This system is important because it:
- Reduces manual errors in academic management
- Improves transparency in student records
- Ensures fairness in grade release
- Strengthens communication between departments
- Protects sensitive academic data
- Encourages digital transformation in schools

11. CHALLENGES ADDRESSED

Manual record keeping issues

Delayed grade release

Unauthorized access to results

Lack of coordination between staff

Payment tracking difficulties

12. FUTURE IMPROVEMENTS

The system can be improved by adding:

Mobile application version

Email and SMS notifications

Online payment integration

AI-based academic performance tracking

Biometric login security

Advanced analytics dashboard

SUSTAINABILITY (REPORT SECTION)

The sustainability of the School Student Portal System refers to its ability to continue operating effectively over a long period while remaining useful, adaptable, and relevant to future needs. This system is designed in a way that it can grow and improve as school requirements change. It is not a fixed system, but one that can be expanded with new features and technologies.

Key Points of Sustainability:

- The system is built using a flexible structure that allows future improvements.
- It can support new features such as mobile applications, online payments, and notification systems.
- It can be adapted to handle more students, lecturers, and administrators as the institution grows.
- The system supports digital transformation in education, reducing reliance on manual processes.
- It encourages long-term use by improving efficiency and reducing administrative workload.

Long-Term Benefits:

Reduces paper-based record keeping

Supports continuous academic management

Improves consistency in school operations

Enhances digital learning environment

Overall, the system is sustainable because it can evolve with technology and continue to serve educational needs over time.

MAINTENANCE

Maintenance refers to the process of keeping the School Student Portal System updated, functional, and secure after it has been developed and deployed. Regular maintenance ensures that the system continues to work properly without errors and remains safe for users.

Types of Maintenance:

1. Corrective Maintenance

Fixing errors or bugs in the system

Resolving login or data issues

Correcting system failures when they occur

2. Preventive Maintenance

Updating the system regularly to avoid future problems

Monitoring system performance

Strengthening security features

3. Adaptive Maintenance

Updating the system to work with new technologies

Modifying the system to support new requirements (e.g., new grading system or payment method)

Maintenance Activities Include:

Regular software updates

Database backup and recovery

Security checks and user access control review

Fixing broken features or errors

Improving system performance and speed

Importance of Maintenance:

- Ensures system reliability and stability
- Prevents data loss and security breaches
- Improves user experience for students, lecturers, and administrators
- Keeps the system up to date with modern technology

Conclusion on Maintenance: Proper maintenance ensures that the School Student Portal System remains efficient, secure, and fully functional over time. It helps the system stay useful and dependable for continuous academic operations.

13. CONCLUSION

The School Student Portal System is more than just a school project. It represents a real solution to real educational challenges. By combining administration, academic evaluation, and student access into one controlled system, it creates fairness, structure, and efficiency. This project shows how technology can improve education and reduce stress for students, lecturers, and administrators. In the future, systems like this will continue to shape how schools operate, making learning environments more digital, transparent, and reliable.

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